ENGINEERING CASEBOOK · 001

Water, Creep & Very Small Margins

Issue 001 · original cases · reconstructed 18 Aug 2026

ARCHIVAL RECONSTRUCTION

DEEP DIVE · CHAMPLAIN TOWERS SOUTH

punching-shear cone / load redistribution

YOU ARE THE ENGINEER
A roughly 40-year-old reinforced-concrete condominium near the sea has a flat pool deck supported directly on columns. Long-term water exposure, reinforcement corrosion and construction deviations have reduced reserve capacity. One slab-column connection begins to distress. What concerns you more: the local damage, or where the redistributed load will go next?

NIST's 2026 technical findings concluded that two garage-column-to-pool-deck connections suffered punching-shear failure in early June 2021, about three weeks before the catastrophic collapse on 24 June. Cracking then progressed, loads redistributed to adjacent connections with inadequate reserve, and the pool deck ultimately broke away and damaged additional tower-support connections.

The useful lesson is not merely “check punching shear.” It is to ask what happens after one element stops carrying the load your model assigned to it. A structure that technically passes with very little reserve is not equivalent to one with substantial redundancy once deterioration, construction tolerances and modifications enter the picture.

DOCUMENTED FINDING

NIST reported narrow margins against failure caused by design/code issues and construction deviations, with later modifications and long-term corrosion further reducing capacity. The investigation also ruled out several competing explanations, including foundation failure and nearby-construction vibration.

ENGINEER'S NOTEBOOK

When reviewing an existing flat slab, do not stop at “does this connection calculate?” Also ask: if this connection degrades or fails, where does the load go next, and can the adjacent system carry it?

SOURCES FOR THIS CASE

NIST · Technical findings on Champlain Towers South collapse · 2026
https://www.nist.gov/news-events/news/2026/06/nist-releases-technical-findings-what-caused-2021-partial-collapse

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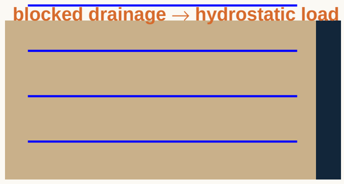
Problems from Practice

Retaining-wall drainage + adhesive-anchor creep

SITE PROBLEM · THE FLOOR SCREED WAS TRYING TO SEND A DISTRESS SIGNAL



A multi-storey steel-frame building sat beside a retaining wall on a stepped site. The first obvious symptom was almost insultingly mundane: corridor screed began popping upward. Investigation found the retaining wall itself was moving.

The wall had been designed assuming granular, fully drained fill. In service, the retained material behaved as water-bearing silty ground. The drainage arrangement included a flexible 100 mm plastic pipe that lacked proper falls, could not readily be rodded and had become silted.

WHY IT MATTERED

The moving retaining wall was transferring lateral action into the steel frame, whose stability bracing had never been intended to restrain a saturated hillside. The remedy combined improved drainage, a waling truss, a deliberate movement zone and monitoring: water management plus structural restraint plus movement accommodation.

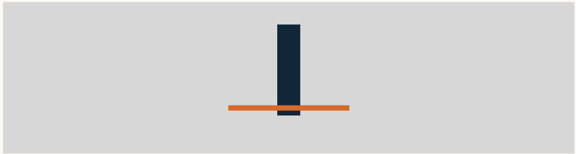
FIELD TRIGGER

On retaining-wall inspections, look beyond the wall itself: blocked outlets, standing water, settlement behind the wall, cracked finishes, displaced paving and unexpected contact with nearby structures can all reveal the true load path.

SOURCE

CROSS-UK · Poor retaining wall design threatens stability · Report 1420
https://www.cross-safety.org/uk/safety-information/cross-safety-report/poor-retaining-wall-design-threatens-stability-1420

DETAIL / MATERIAL · THE EPOXY WAS STRONG. THAT WASN'T THE QUESTION.



sustained tension + creep

In Boston's I-90 Connector Tunnel, a suspended concrete ceiling was supported using steel hardware anchored overhead with **Powers Power-Fast Epoxy Injection Gel, Fast Set formulation**, also supplied as NRC-1000 Gold. On 10 July 2006, roughly 26 tons of panels and support hardware fell onto a vehicle, killing one occupant.

NTSB identified inadequate resistance to **creep under sustained tensile load** as the probable cause. Short-duration strength was not enough. Post-accident FHWA testing showed Fast Set and Standard Set behaved similarly in short tests but radically differently under sustained loading.

THE PRODUCT DECISION

The manufacturer's Standard Set formulation had undergone creep qualification testing; the Fast Set formulation had not demonstrated equivalent long-term performance. The failure was therefore not a generic “chemical anchors are bad” story. It was a compatibility failure between the exact formulation and the load-duration demand.

SPECIFICATION LESSON

For safety-critical adhesive anchors, specify the exact qualified system, substrate, embedment, hole-cleaning procedure, installation temperature and load duration. “Approved chemical anchor” is not a serious sustained-load specification.

SOURCE

NTSB · Ceiling Collapse in the Interstate 90 Connector Tunnel, Boston · HAR-07/02
https://www.ntsb.gov/investigations/AccidentReports/Reports/HAR0702.pdf

Engineering Done Well

Leiden wet excavation · issue synthesis

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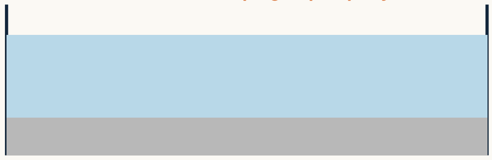
ENGINEERING WIN · BUILD THE GARAGE WITHOUT FIRST MAKING THE HOLE DRY

excavate wet → plug → pump dry



A five-level underground parking garage in historic Leiden required an excavation about 18 m deep with groundwater only about 0.6 m below ground level and soft silty sands, clay and peat above stronger sand. Nearby historic buildings made uncontrolled drawdown unattractive.

Instead of trying to keep a deep permeable excavation dry throughout construction, engineers excavated **wet**. Diaphragm walls and staged restraint controlled the excavation. At final depth, an underwater concrete floor was placed and tied into anchor piles/barrettes so that the system could resist uplift once the water inside was pumped out.

Only after the retaining system and impermeable bottom were established was the excavation drained and the permanent slabs built. Monitoring compared diaphragm-wall deflections and surrounding settlement with predictions, and the observations were then used for back-analysis.

THE TOOLBOX

The documented systems were diaphragm walls, staged steel strutting, wet excavation, underwater concrete, anchor piles/barrettes, instrumentation and progressive replacement of temporary restraint by permanent slabs.

WHY IT WORKED

The construction sequence changed the hydraulic problem rather than attempting to overpower it. Maintaining water inside the excavation limited the hydraulic-head difference during excavation and avoided asking a giant dewatering system to solve every risk simultaneously.

SOURCE

ISSMGE · Design, Construction and Back Analysis of a Deep Underground Parking Garage in an Urban Environment
https://geocasehistoriesjournal.org/issues/volume-7-1/design-construction-and-back-analysis-of-a-deep-underground-parking-garage-in-an-urban-environment/pdf

THE THREAD · DESIGN ASSUMPTIONS HAVE SERVICE LIVES TOO

Champlain: reserve capacity and load redistribution mattered as deterioration accumulated. **Retaining wall:** “fully drained” ceased to be true when the drainage system deteriorated. **Boston:** short-term adhesive strength said nothing about long-term creep. **Leiden:** the team challenged the assumption that the excavation had to be dry before construction could proceed.

60-SECOND TAKEAWAY · Local failure is not necessarily local. Drainage is a maintained condition, not a permanent property. Proprietary-product qualification must match the real load case. Construction methodology can eliminate a geotechnical problem more elegantly than brute-force capacity.